ICSMMEP-26 — Konuşma Metni

Eggshell–Agar Biocomposites: Development and characterisation for sustainable interior applications Betül Ünal · Cappadocia University · Mecca, 3–4 November 2026

Nasıl kullanılır

Kalın yazılan cümleler slaytta birebir duruyor. İstersen ekrandan okursun, istersen izleyiciye söylersin. Düz yazılanlar yalnız bu metinde.

Köşeli parantez içindeki Türkçe notlar sahne yönergesidir, okunmaz. Sayıları yazıyla yazdım; kürsüde rakam görüp duraksamayasın diye.

Metin	**11:15** · 140 kelime/dakika
26 slayt geçişi, duraklar	~1:15
Sahnede	**≈ 12:30**

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1 · Cover — 0:15

Good morning. My name is Betül Ünal, from the Department of Interior Architecture at Cappadocia University. The panel in the photograph is what this paper is about: a composite made from waste eggshell, bound in agar.

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2 · Background — 0:30

Eggshell is produced continuously as a by-product of food processing, and almost all of it goes to landfill. Chemically it is very close to pure calcium carbonate, and its specific gravity is lower than that of most mineral fillers used in construction.

[Sağdaki şemaya işaret et]

The case for eggshell as a mineral phase is settled. This paper is about the phase that binds it.

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3 · Prior applications — 0:35

This table is the literature the study departs from. Eggshell has already been used in concrete, in brick

and clay stabilisation, in asphalt, in ceramic tile, in PVC and polyurethane foam, and even in bone scaffolds.

[Elini orta sütuna götür, aşağı indir]

Read the middle column downwards. In every one of these cases the mineral is recovered from waste, but the binding phase remains cementitious, ceramic or petrochemical. **The burden is carried by the binder, and the binder is left untouched.**

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4 · Research gap — 0:35

Three gaps follow from that. The first is that the binding phase is never biological: **no study reports an eggshell composite in which both phases are of biological origin.** The second is that agar has been characterised only as an unfilled film in tension, and **never as the binder of a mineral-filled solid in bending.** The third is that **property data is scattered across separate studies on separate formulations.**

This is not a novel constituent. It is a documented protocol and a single coherent property set.

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5 · Aim — 0:20

The aim is deliberately narrow: **to document a reproducible production protocol for a fully bio-based eggshell–agar composite, and to characterise it against published test standards.** Of the four objectives, **the fourth, deriving an evaluation criterion for this material class, is where I will spend most of my time.**

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6 · Scope — 0:20

I want to be explicit about the boundaries of this work, so that nothing I say is read beyond them.

[Alttaki şeride işaret et]

The filled boxes are the properties I determined. The outlined boxes are the ones I did not, and those include durability, fire classification and life-cycle assessment.

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7 · Method — 0:20

The method has four stages, and the order matters. I produced the material and recorded every parameter, measured it against published standards, positioned it against two material families, and only then proposed a criterion for evaluating it. **The properties came first, and the application envelope was derived from them.**

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8 · Research questions — 0:20

There are five questions. The first four are about formulation and benchmarking, but the fifth is the one I care about: whether compressive strength is an adequate evaluation criterion for this class of material at all. **The first four I can answer by measurement. The fifth I can only answer by argument.**

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9 · Hypothesis — 0:25

The hypothesis reads as follows. **A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance as an interior surface material is governed by deformation capacity rather than by compressive strength.** Those are two separate claims, and different parts of the paper test them.

The dissertation behind this paper frames a broader hypothesis around spatial resilience, and what you see here is its materials-engineering component.

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10 · Stage 1 · Mineral phase — 0:30

The first stage prepares the mineral. The shell comes from a single bakery, and it is cleaned by hand, heat-treated at one hundred and twenty degrees for twenty minutes, milled, and sieved. **I chose one hundred and fifty micrometres so that the mineral phase here would be comparable with the eggshell polymer-composite literature. The heat step is decontamination and drying rather than sterilisation, because sterility was never verified.**

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11 · Stage 2 · Binder preparation — 0:40

The second stage prepares the binder, and I have drawn it as a temperature–time profile because that is what it is.

Agar is thermoreversible: it melts when you heat it and sets again as it cools, so no hardener and no cure temperature are needed. The agar is added a little at a time under constant stirring, because added all at once it forms agglomerates that never redisperse. It is held at ninety-five degrees until fully dissolved, brought down to eighty while the viscosity builds, and the shell goes in near the end, and the whole preparation takes fifty-five minutes.

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12 · Stage 3 · Casting and cure — 0:30

The third stage is casting and curing. The mixture is cast into silicone moulds, demoulded on the fourth day, and left for another seven days at around twenty-three degrees. **After the fifty-five minute preparation the material receives no further heat, so the entire cure happens at room temperature.**

[Panele işaret et]

The composite shrinks as the water evaporates, and I did not log the dried dimensions, so every dimension I quote today is a mould dimension.

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13 · Formulations — 0:30

Two things to notice in this matrix. In the compression series the shell-to-binder ratio is exactly two in all three mixes, which means that **solids concentration is the only variable, and you can see it in the bars: the water band gets shorter from C1 to C3.**

In the flexural series I have to declare a confound. **The glycerol was halved while the batch mass stayed the same, so the water rose to compensate.**

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14 · Test programme — 0:25

Here are the four specimen geometries, with the standard and the number of specimens underneath each one. Before I show you any numbers I would like the reporting convention on the record. **The mechanical values are means of three specimens, and because I did not retain the individual values, I cannot give you standard deviations. The fire, absorption and density figures are single determinations.**

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15 · Compression — 0:30

This answers the first question. Compressive stress rises linearly with solids concentration, from two and a half to four megapascals. Three points will fit almost any line, so I call this a hypothesis rather than a law. But the physical reading is clear. **Strength here is governed by packing density, not by mineral content. The way to a stronger composite is to take water out, not to put more shell in.**

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16 · Compressive failure — 0:30

The curve is linear to about one megapascal, and then the stiffness falls away as the agar network yields and the mineral redistributes. The peak arrives at around ten per cent strain, with no sudden drop in load.

[Fotoğrafı göster]

Look at the cube: it has barrelled, it is shorter and wider, and it is still a single body. Failure proceeds by progressive crushing rather than by fracture. The strain scale assumes ten per cent deformation from observation, not instrumented measurement.

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17 · Flexure — 0:30

In bending the decisive variable is the plasticiser. At eighteen point eight per cent glycerol the specimen deformed without developing any measurable resistance; halving the glycerol gave two point three megapascals at a modulus of one hundred and ninety-two. **The transition is sharp, and it lies between those two compositions.**

The specimen is kerfed on its tension face while the standard assumes a solid rectangle, so the modulus I am quoting is a lower bound.

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18 · Flexural deformation — 0:20

I would rather show you this than ask you to take it from the numbers. **The beam is deflected well past the point where a comparable ceramic would have parted, and it is still intact. Both test programmes gave me the same result: substantial deformation with the integrity retained.**

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19 · Fire, water, density — 0:30

The material ignites at around ten seconds under a direct flame and extinguishes itself four seconds after the flame is taken away, with no lateral spread and no dripping. **It is an ignitability test, and there is no Euroclass classification behind it.**

The mass gain over twenty-four hours of immersion was zero point seven per cent, so **it is the coating that limits humid service rather than the composite itself.**

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20 · Conventional materials — 0:25

This is the first benchmark, against the materials it would sit beside in an interior. **It has one sixth the compressive strength of MDF and one twentieth that of travertine. On strength alone this composite is not competitive with engineered wood or stone, and I am not going to claim that it is.**

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21 · Bio-based composites — 0:30

The second benchmark is the one that matters. In compression it stands an order of magnitude above mycelium and hempcrete, and in stiffness, at one hundred and ninety-two megapascals, it sits inside the LDPE band.

[Logaritmik eksene işaret et]

It was cast from water and dried at room temperature, with no oven, no autoclave and no extruder, and it still reaches the stiffness of melt-processed bio-polymers.

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22 · Modulus of resilience — 0:55

This brings me to the fifth question. The diagram is the argument.

The shaded area is the energy a material absorbs per unit volume before permanent damage. That area is one half sigma epsilon, and since epsilon is sigma over E, it reduces to sigma squared over two E. **Notice where the modulus sits in that expression: it is in the denominator.** So the low stiffness that counted against the material is exactly what raises this quantity: about fourteen kilojoules per cubic metre.

Strength tells you the load at which a material stops performing. It does not tell you whether it then fails gracefully or catastrophically, and for a demountable component that is what decides service life.

I cannot give you a comparative table, because the literature reports strength and modulus in separate studies.

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23 · Application envelope — 0:20

The application envelope follows from the measurements: non-loadbearing surfaces, dry interiors, away from ignition sources, and modular, so that damage can be replaced one unit at a time. **I did not carry out a life-cycle assessment, so these are qualitative potentials rather than quantified claims.**

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24 · Limitations and further work — 0:25

I have put each limitation next to the experiment that resolves it, because that pairing is the point. There is no dispersion data, the strain was assumed rather than instrumented, and the shrinkage was never quantified. **Every value I have given you today is a lower bound, and I would rather set that out myself than have you find it.**

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25 · Conclusions — 0:25

Let me close with five points. The protocol can be repeated. Packing density, not mineral content, governs compressive strength. Plasticiser content decides whether the material is brittle or ductile. A water-cast, ambient-dried material can reach bio-polymer stiffness. **And strength on its own is not an adequate criterion for this class: strength and modulus should be reported together.**

Thank you. I am happy to take questions.

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26 · References — 0:00

[Slaytı açık bırak, bir şey söyleme. Soru gelirse kaynağı buradan göster.]

Soru–cevap

Kısa tut. Sınırlılığı kabul et, sonraki deneyi söyle, sus.

"Standart sapma yok, nasıl güveneceğiz?" You are right, and I said so before I showed any numbers.

There were three specimens per formulation, and the individual values were not retained. Repeating the series with dispersion reported is the first row of my limitations table.

"Basma testi hangi standarda göre?" It was cube compression rather than a named standard, and I flag that as a deviation. The flexural work follows ASTM D790-17, the ignitability test ISO 11925-2, and the absorption test ASTM D570.

"Ölçüler kalıp ölçüsü, gerçek gerilme değil." That is correct. The composite shrinks as the water evaporates and I did not log the dried dimensions, so every value I reported is a lower bound. Recording the dried geometry is in the limitations table.

"Yivli numuneden modül hesaplanmaz." I agree, and that is why I called it a lower bound rather than a material constant. Testing an un-kerfed section is in the table.

"R^2 = 0.99 ama üç nokta var." That is exactly why I called it a hypothesis rather than a law. What I am claiming is the physical reading: packing density, not mineral content, governs strength.

"Neden agar? Nişasta ya da jelatin olmaz mıydı?" Agar sets thermoreversibly without a hardener, and it carried sixty-six per cent mineral loading without a coupling agent. Comparing binders would be a separate study, and it is a fair thing to ask for.

"Yangın mekanizması nedir?" The probable mechanism is endothermic decomposition of the carbonate fraction, which dilutes the flame front. I say probable because it was not verified calorimetrically.

"Nem, küf, yaşlanma?" All untested, and named as such on the limitations slide. Durability of any kind is outside the scope of this paper.

"Ölçeklenebilir mi?" I have no data on that. What I can say is that the protocol needs a hotplate, a mould and ambient drying, and nothing else.

"Yangın sınıflandırması olmadan iç mekanda nasıl kullanılacak?" It cannot be specified without one, and I say so on the application slide. What I report is ignitability behaviour, which is the precondition for seeking a classification rather than a substitute for it.

Provaya dair

- Yüksek sesle, kronometreyle oku. Metnin kendisi 11:15; sahnede duraklarla

12:30 civarı.

- dakikayı geçersen **hızlanarak değil, cümle atarak kısalt.** Sırasıyla:

- Slayt 16, *"The strain scale assumes ten per cent deformation..."* cümlesi —

soru gelirse söylersin.

- Slayt 10, *"The heat step is decontamination and drying..."* cümlesi.
- Slayt 22, *"I cannot give you a comparative table..."* cümlesi.
- Slayt 13, ikinci paragraf (eğilme serisindeki confound) — slaytta yazıyor.
- Dokunulmayacak slaytlar: **3, 4, 11, 13, 15, 17, 22.** İskelet bunlar.
- Köşeli parantezli yönergelerde gerçekten ekrana dön ve işaret et.